

TIC 168789840: Eclipsing Binary Classification

Finding: TIC 168789840

Mission: TESS

TIC 168789840 is a complex stellar system containing three eclipsing binaries. The Lightcurve-based classifier was applied to its TESS photometric data to test the performance of the period detection and classification pipeline.

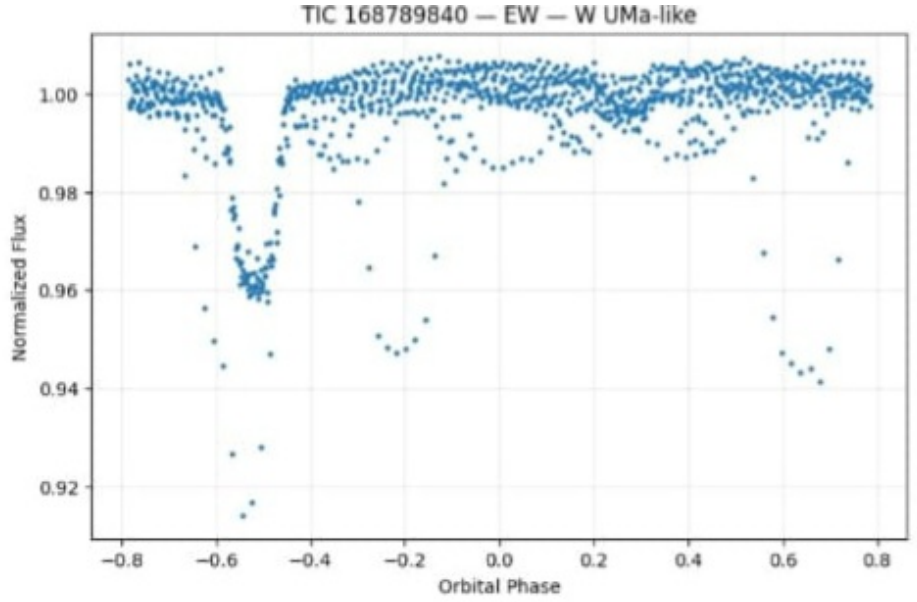


Figure 1: Light curve and analysis output for TIC 168789840.

The program identified an orbital period of

$$P = 1.5705 \text{ days},$$

which closely corresponds to the period of one of the eclipsing binaries within the system. This indicates that the period-search algorithm successfully detected a genuine periodic signal.

The measured primary and secondary eclipse depths were

$$D_{\text{primary}} = 0.0586, \quad D_{\text{secondary}} = 0.0528.$$

The eclipse-depth ratio was calculated as

$$R = \frac{D_{\text{secondary}}}{D_{\text{primary}}} = \frac{0.0528}{0.0586} \approx 0.901.$$

Since the two eclipse depths are relatively similar, the rule-based classifier classified the detected light-curve morphology as

EW – W UMa-like

The folded light curve also contains additional variability and scattered features. This is expected because TIC 168789840 is a complex multiple-star system. The observed TESS flux contains contributions from its three eclipsing binaries, meaning that signals from the other components can also appear in the combined light curve.

Therefore, the EW classification should be interpreted as the classification of the detected **1.5705-day signal**, rather than a classification of the entire system.

Conclusion: The result demonstrates that the program can recover a genuine binary period even from a complex multiple-star system. It also highlights a limitation of the current single-period, rule-based classifier, since multiple binary components can contribute to the observed light curve.